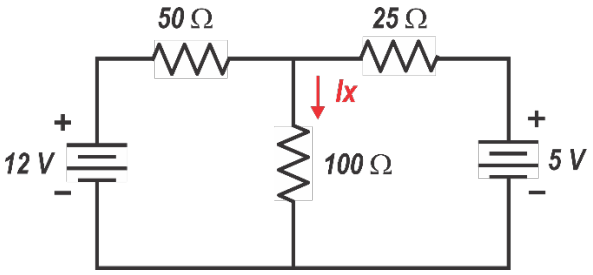
1) Mesh Analysis: Use mesh analysis to answer the following questions.



A) Assign mesh currents to this circuit. Redraw the circuit and clearly show the mesh currents including the assumed current rotation (clockwise or counterclockwise).

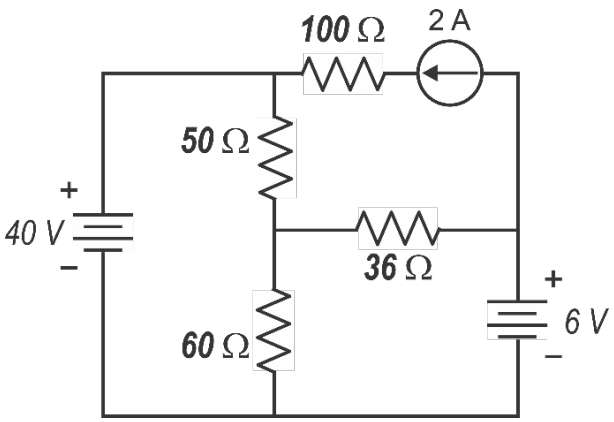
B) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

C) Using KVL around the loop for each of your mesh currents, write a sufficient set of equations to solve for the unknown mesh currents. Clearly list your complete set of equations.

D) Solve the equations in (C) for the unknown mesh currents and I_X. Calculate the unknown currents by hand, do not use a software solver. Provide a **Table** of the mesh current values and I_X.

E) Find the **Power** in the two ***voltage*** sources. For each, determine whether the source is delivering or absorbing power, and briefly explain why.

2) Mesh Analysis: Use mesh analysis to answer the following questions.



A) Assign mesh currents to this circuit. Redraw the circuit and clearly show the mesh currents including the assumed current rotation (clockwise or counterclockwise).

B) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

C) Using KVL around the loop for each of your mesh currents, write a sufficient set of equations to solve for the unknown mesh currents. Clearly list your complete set of equations.

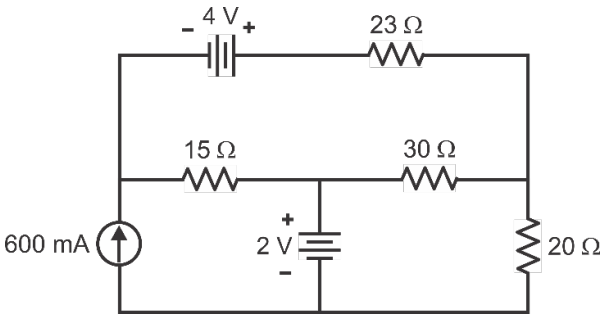
D) Solve the equations in (C) for the unknown mesh currents. Calculate the mesh current by hand, do not use a software solver. Provide a **Table** of the mesh current values.

E) Find the **Power** in the two ***voltage*** sources. For each, determine whether the source is delivering or absorbing power, and briefly explain why.

F) Calculate the **power** for the ***current*** source. Determine whether the current source is delivering or absorbing power, and briefly explain why.

Next page for questions 3 and 4

3) Mesh Analysis: Use mesh analysis to answer the following questions.



A) Assign mesh currents to this circuit. Redraw the circuit and clearly show the mesh currents including the assumed current rotation (clockwise or counterclockwise).

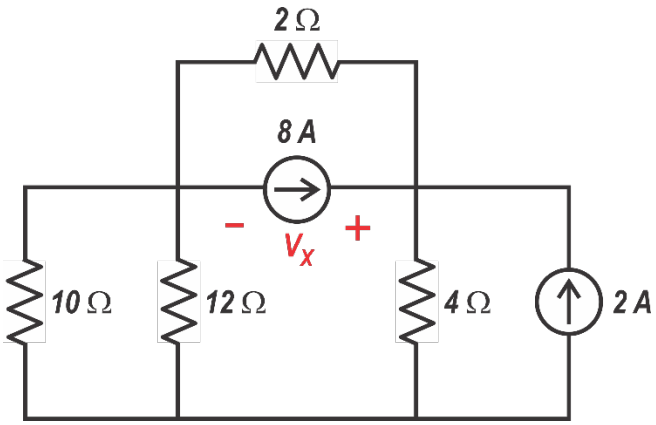
B) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

C) Using KVL around the loop for each of your mesh currents, write a sufficient set of equations to solve for the unknown mesh currents. Clearly list your complete set of equations

D) Solve the equations in (C) for the unknown mesh currents. Calculate the unknown currents by hand, do not use a software solver. Provide a **Table** of the unknown mesh currents.

E) Find the **Power** in the two voltage sources (4 V source and 2 V source). State if the power is *absorbed* or *delivered* for each and why.

4) Mesh Analysis: Use mesh analysis to answer the following questions. Do not use a Supermesh.



A) Assign mesh currents to this circuit. Redraw the circuit and clearly show the mesh currents including the assumed current rotation (clockwise or counterclockwise).

B) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

C) Using KVL around the loop for each of your mesh currents, write a sufficient set of equations to solve for the unknown mesh currents. Clearly list your complete set of equations.

D) Solve the equations in (C) for the unknown mesh currents and V_X. You may use a use a software solver, such as Wolfram-Alpha, www.wolframalpha.com. NYU has a site license which provides full functionality. Provide a **Table** of the unknown mesh currents and V_X.

E) Find the **Power** in the two ***current*** sources. For each, determine whether the source is delivering or absorbing power, and briefly explain why.